

Brawlhalla AI Agent

Reinforcement Learning in Fighting Games



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What is Brawlhalla?

- A fast-paced 2D platform fighting game (similar to Super Smash Bros)
- Players aim to knock opponents off the stage
- Requires:
 - Fast reaction time
 - Precise movement
 - Strategic decision-making
- Why it's hard for AI?
 - No game state API. The agent sees only raw pixels, just like a human would

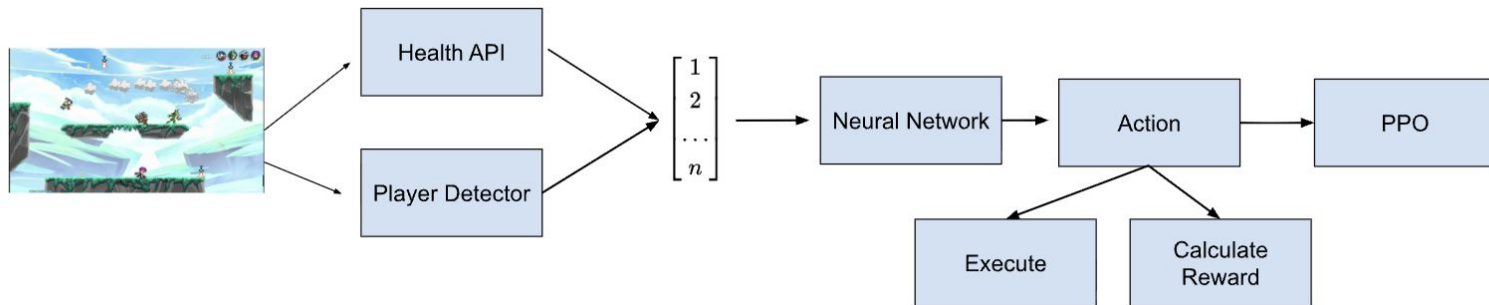
Our Goal

- Build an AI Agent that can play Brawlhalla autonomously
- Learn gameplay directly from raw screen input
- Develop:
 - Movement control
 - Attack timing
 - Strategy and adaptation
- Target: Live gameplay against CPU or real player

Why does this matter?

- Brawlhalla is genuinely a hard RL problem:
 - Split second decisions
 - Sparse rewards
- Directly relevant to the core course themes:
 - Reinforcement Learning
 - CNNs
 - Real world environment design
- We chose this because we wanted to test how far RL can go in a real game

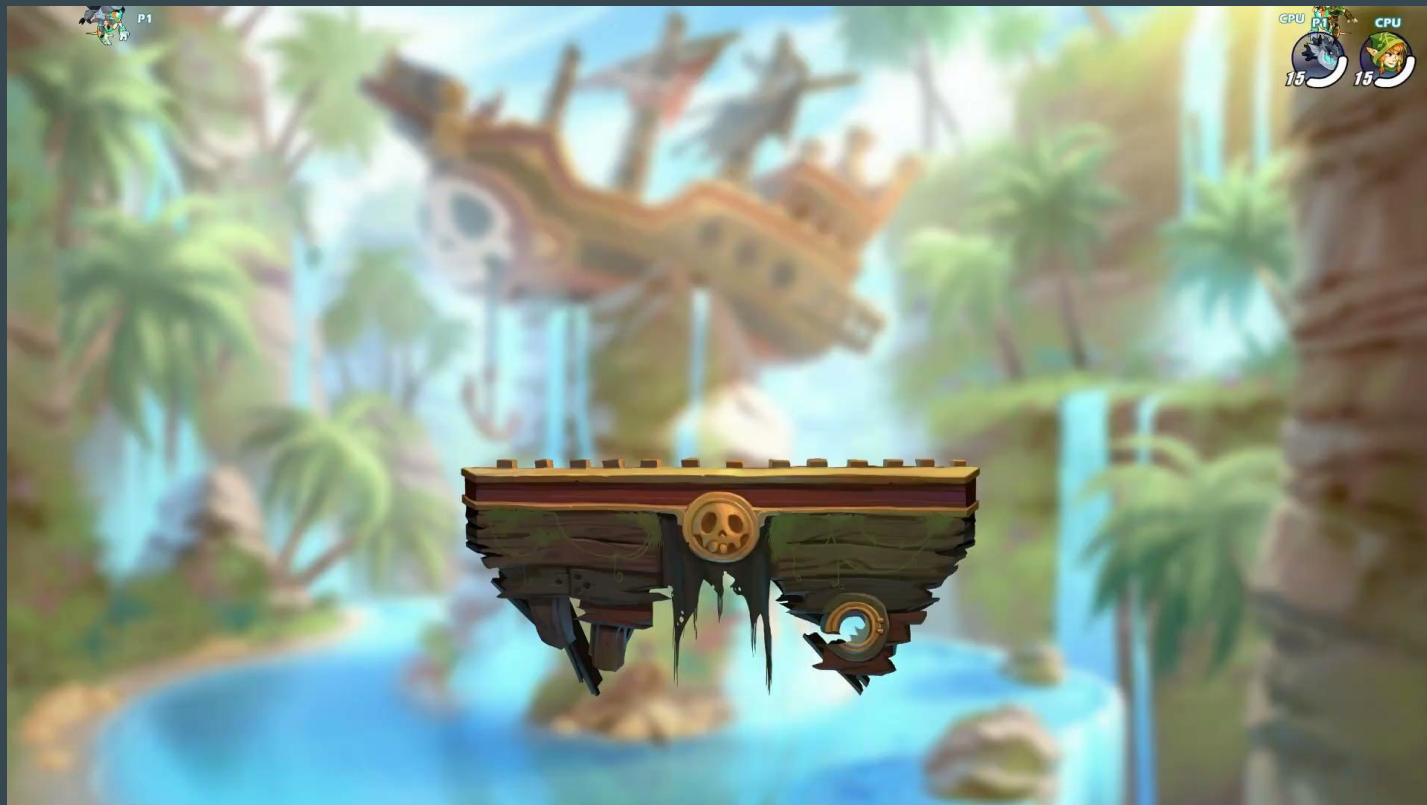
Methodology



Different Neural Network Inputs:

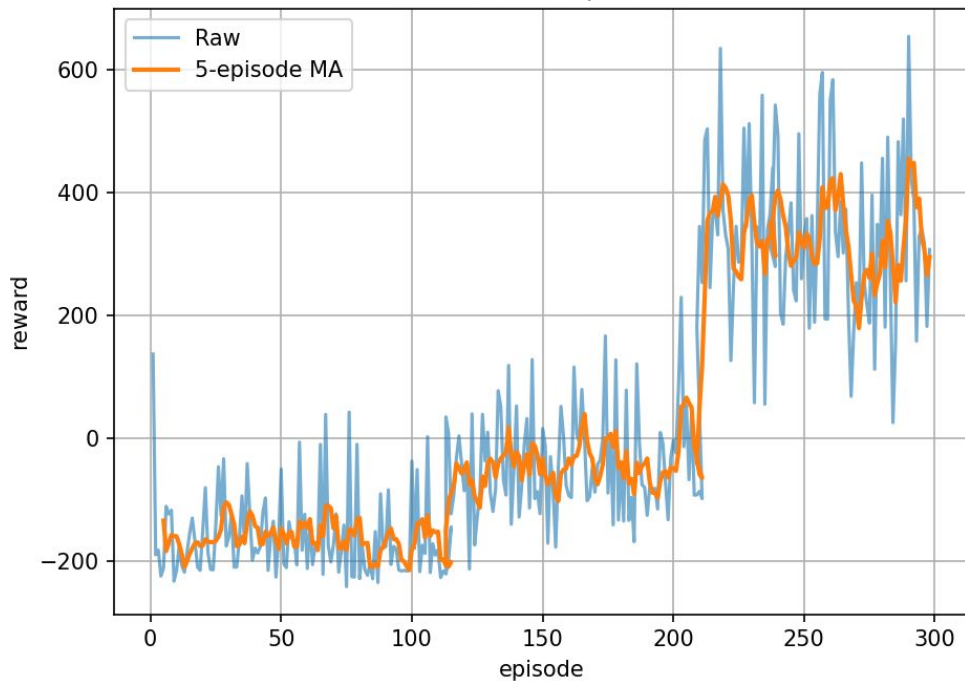
- CNN/APIs info
- Solo APIs info
- Behavioral Cloning/APIs info

Gameplay



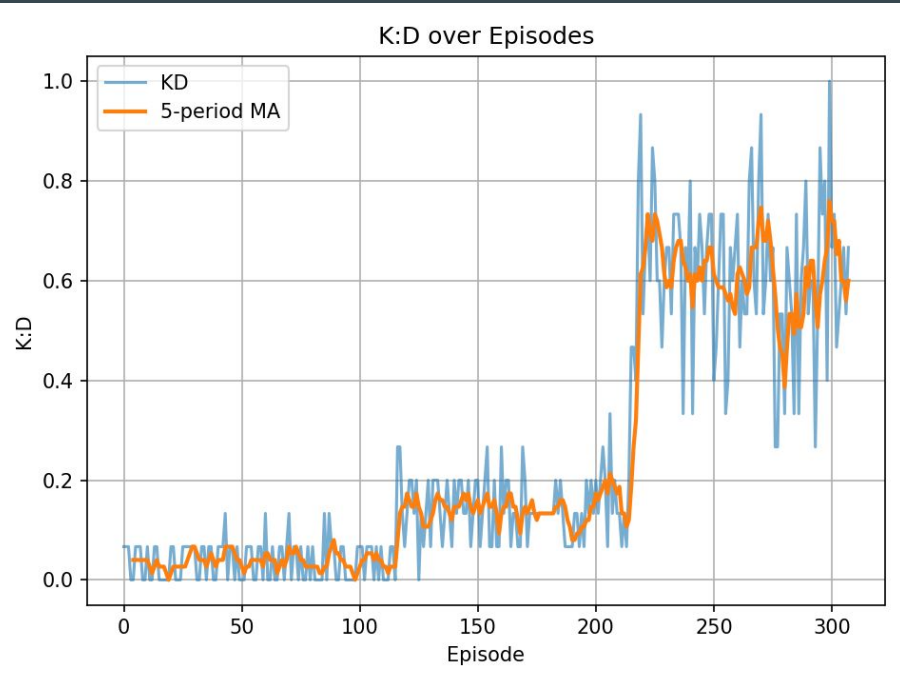
Results: Reward

Reward over Episodes



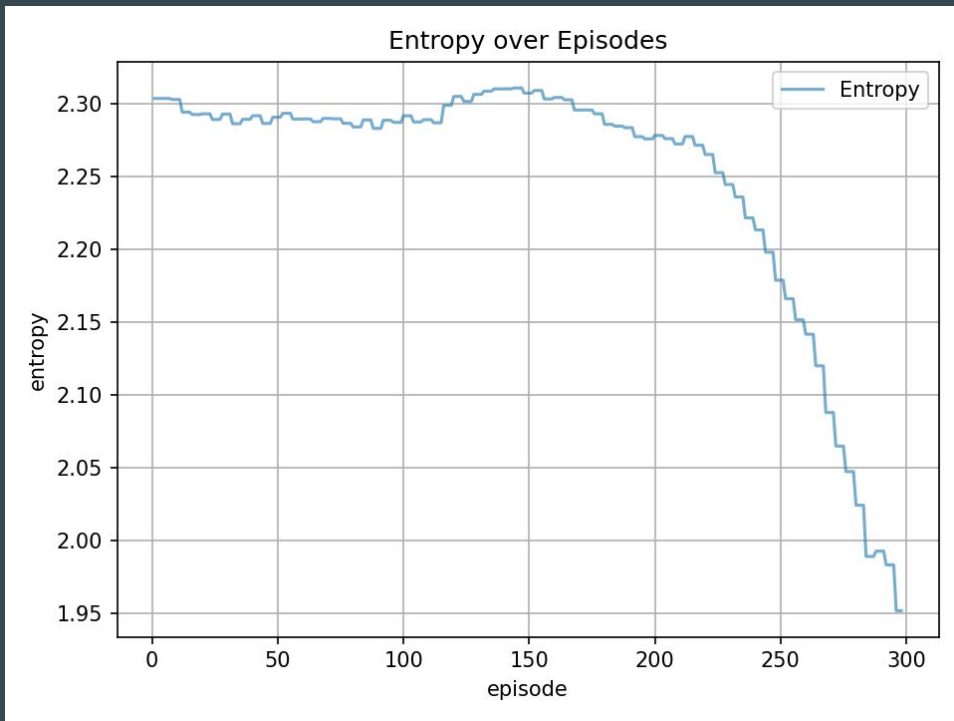
- Our agent was able to shape an average reward of 24.18 over 298 episodes
 - 36 hours of training and 4470 agent deaths
- Showed steady improvement, with some occasional expected variance

Results: K:D



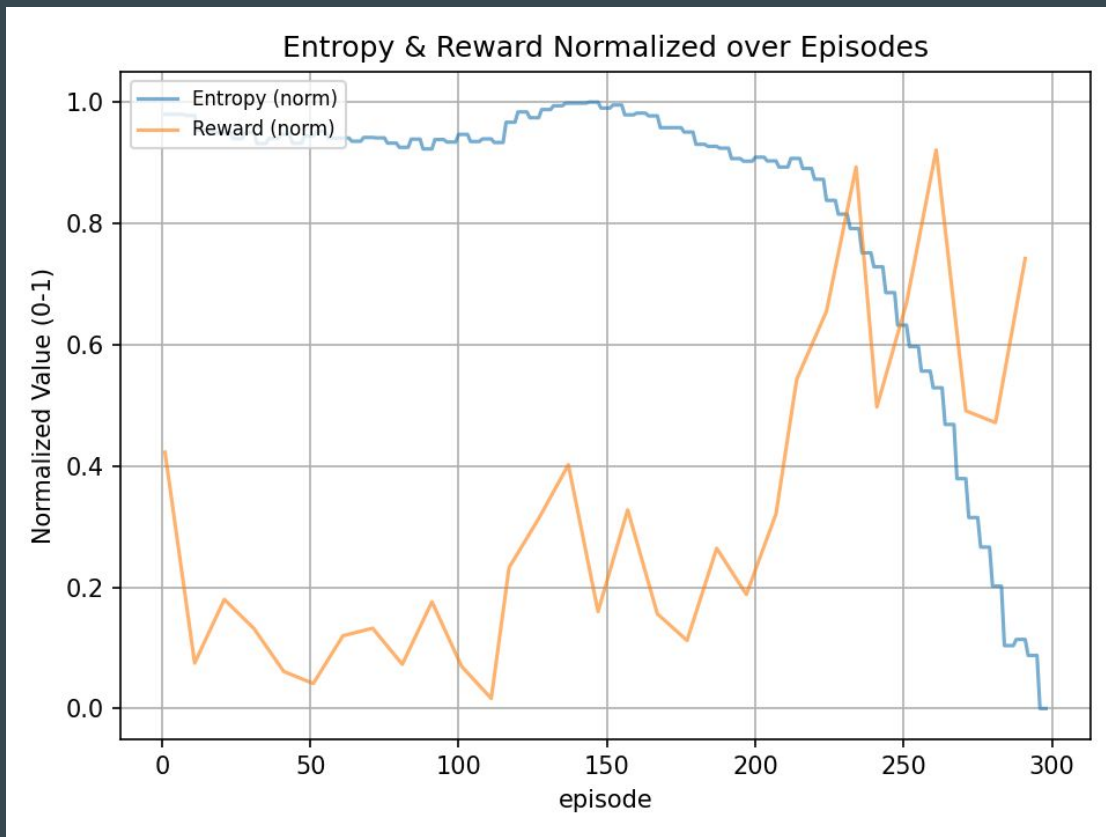
- K:D is a common metric for performance in video games
 - $(\text{Your Kills}) / (\text{Your Deaths})$
- Our agent was able to shape an average K/D ratio of about 0.24
 - In comparison, a random agent had a K/D of about 0.03

Results: Entropy

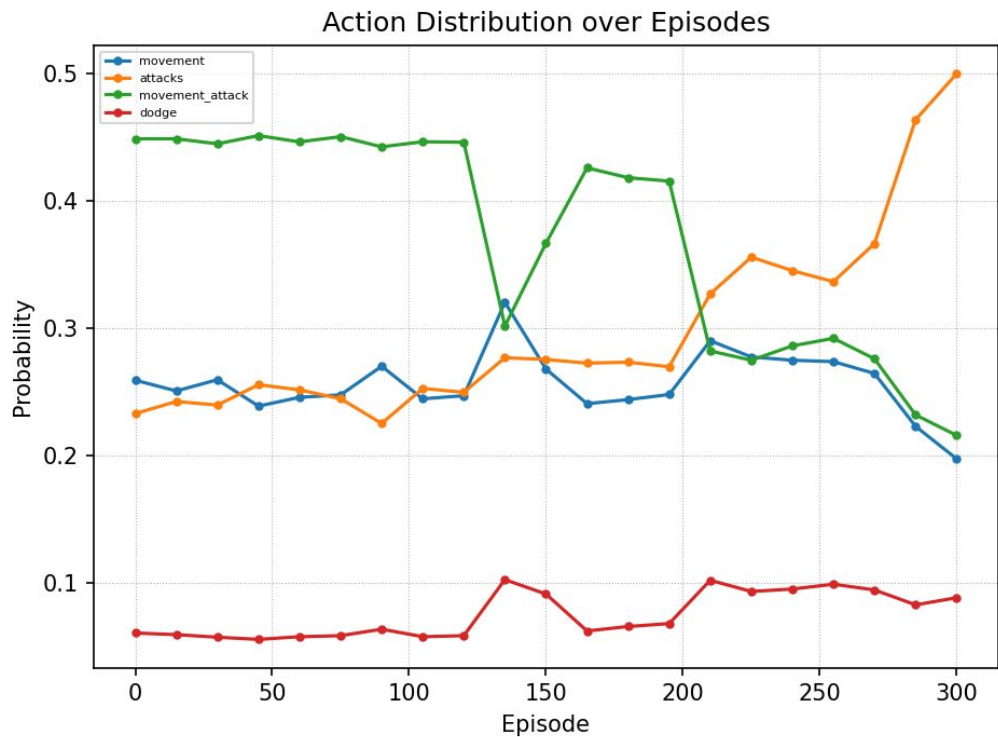


- Entropy is a measure of how diverse the probability distribution is amongst actions
 - Max entropy 11 actions = $\ln(11) = 2.398$
- The more confident the agent starts to get, the lower the entropy
 - Agent starts to prefer one action, probability for action becomes higher, and converges on that action
- Don't want full entropy collapse!
 - Fighter games are not won by performing one action repeatedly!

Results: Entropy and Reward



Results: Action Distribution



- 11 actions grouped into 4 categories
- Towards latter episodes, agent preferred stationary attack actions
 - Led to less accidental suicides, falling off stage, while also dealing damage and earning positive reward

Takeaways

- Our agent was able to shape a K/D ratio of about 0.24
 - In comparison, a random agent had a K/D of about 0.03
 - 8x improvement over random agent
- Reward shaping is HARD
 - Agent often exploits reward functions
 - Lots of noise
 - Simpler the better
- Real-time environments create training bottlenecks
 - Cannot be parallelized or sped up
- Having huge input layers to NN may just be too much noise

Future Steps

- Let our agent keep training!
 - How will our agent perform against higher difficulty CPUs?
- Refine the reward system to find most optimal reward function
- See how our model performs with different characters
 - Movement speed, attack damage, shield, differ amongst characters